CHAPTER 11 – Matrix Inverse and Condition

$$A A^{-1} = A^{-1} A = I \quad , \quad A: \text{square matrix}$$

Calculating the inverse

- Compute in a column-by-column way:

  if $b = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ ⇝ solution to $Ax=b$ will be the first column of A^{-1}.

  $x = A^{-1} b \hookrightarrow \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

  if $b = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ ⇝ sol^n to $Ax=b$ is $x = A^{-1} b \Rightarrow 2^{nd}$ column of A^{-1}.

- Can use LU factorization to solve for A^{-1}: $(A = LU)$

1st column of A^{-1}: $\underbrace{LU}_{d}\, x = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \Rightarrow Ld = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \xrightarrow{\text{solve}} d \Rightarrow Ux = d \xrightarrow{\text{Solve}} x$

2nd col. of A^{-1}: $\underbrace{LU}_{d}\, x = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \Rightarrow Ld = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \rightarrow d \Rightarrow Ux = d \xrightarrow{\text{Solve}} x$

3rd col. of A^{-1}: $LUx = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \Rightarrow Ld = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \rightarrow d \Rightarrow Ux = d \xrightarrow{\text{Solve}} x$

Error Analysis and System Condition

Inverse provides a means to discern whether systems are ill-conditioned. (System: $\underline{\underline{A}}\,\underline{x} = \underline{b}$)

(1) Scale A so that largest element in each row is 1. Invert scaled A. If some elements of $\bar{A}^{-1}$ are several orders of magnitude greater than 1, it is likely that system is ill-conditioned. (More explanation will come later).

(2) Multiply the inverse by the original coefficient matrix and see whether the result is close to identity. If not ⇒ ill-conditioned.

(3) Invert the inverted matrix and see if it is sufficiently close to the original matrix. If not ⇒ ill-conditioned.

Vector and Matrix Norms

n-dimensional vector: $X = [x_1\ x_2 \cdots x_n]$

Euclidean norm: $\|X\|_e = \sqrt{\sum_{i=1}^{n} x_i^2}$

Matrix A:

Frobenius norm: $\|A\|_f = \sqrt{\sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}^2}$

- A single value that quantifies the "size" of A.

p-norms (vectors): $\|X\|_p = \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p}$ $\rightsquigarrow$ $p=2 \rightarrow$ Euclidean norm (L_2 norm), $p=1 \rightarrow L_1$ norm

$$\|X\|_1 = \sum_{i=1}^{n} |x_i|$$

$$\|X\|_\infty = \max_{1 \le i \le n} |x_i|$$

matrices ($A_{n\times n}$ matrix)

$\|A\|_1 = \max_{1 \le j \le n} \sum_{i=1}^{n} |a_{ij}|$ column-sum norm

$\|A\|_\infty = \max_{1 \le i \le n} \sum_{j=1}^{n} |a_{ij}|$ row-sum norm

spectral norm: $\|A\|_2 = (\mu_{max})^{1/2}$

μ_{max}: largest eigenvalue of A^TA.

Matrix Condition Number

$$\text{Cond}(A) = \|A\| \, \|\vec{A}^{-1}\|$$

↳ matrix condition number

- ill-conditioned if $\text{Cond}(A) \gg 1$

EX $A = \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{bmatrix}$

- 3x3 Hilbert matrix
- use row-sum norm to estimate the condition number
- Note: Hilbert matrix is ill-conditioned.

Soln: Normalize A so that max. element in each row is 1.

$$A = \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ 1 & \frac{2}{3} & \frac{1}{2} \\ 1 & \frac{3}{4} & \frac{3}{5} \end{bmatrix}$$

→ sum each row ⇒ 1.83, 2.166, 2.35.

$$\|A\|_\infty = 2.35$$

$$\vec{A}^{-1} = \begin{bmatrix} 9 & -18 & 10 \\ -36 & 96 & -60 \\ 30 & -90 & 60 \end{bmatrix}$$

2nd row has largest sum

$$\|\vec{A}^{-1}\|_\infty = |-36| + |96| + |-60| = 192$$

$\text{Cond}(A) = (2.35)(192) = 451.2 \gg 1 \Rightarrow$ ill-conditioned.

MATLAB:

```
>> norm(X,p)
>> cond(X,p)    % p: 1, 2, inf, 'fro'
```

↳ built-in functions

Interpreting the Elements of the Matrix Inverse as a Measure of ill-conditioning:

- Scale A so that the largest element in each row is 1.
- Compute A^{-1}
- If Elements of A^{-1} are $>>$ elem. of A $\Rightarrow$ it is likely that system is ill-conditioned.

Insight: System: $AX = B$

Approximate solution: $\hat{X}$

To check if $\hat{X}$ is acceptable, substitute in original eq^n:

$$R = B - A\hat{X} \qquad (1)$$

↳ residual vector

if R is small, we might conclude $\hat{X}$ is acceptable

However, if X is the exact sol^n:

$$0 = B - AX \qquad (2)$$

Subtract (2) from (1):

$$R = A(X - \hat{X})$$

$$A^{-1}R = X - \hat{X} \rightsquigarrow \text{error}$$

↳ if elements of A^{-1} are large, a small R may give a large error

$\Rightarrow$ i.e a small R does not guarantee an accurate sol^n.

(checking a sol^n by substitution may be misleading)

→ if the largest element of $A^{-1} \sim 1$, system can be considered as well-conditioned.

→ if A^{-1} has elements $>> 1$ $\Rightarrow$ system is ill-conditioned

Other ways of finding A^{-1}:

Gauss-Jordan elimination

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & -1 & 2 \\ 0 & 1 & 0 & -1 & 0 & 1 \\ 0 & 0 & 1 & 2 & 1 & -2 \end{array}\right]$$

A $\qquad$ A^{-1}

put the left hand matrix into reduced row echelon form

Using Matrix of Minors and Cofactor Matrix

$$C = \begin{bmatrix} -1 & -2 & 2 \\ 2 & 1 & 1 \\ 3 & 4 & 5 \end{bmatrix} \xrightarrow{(-1)^{i+j}} \begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

Matrix of Minors:

$$\begin{bmatrix} \begin{vmatrix} 1 & 1 \\ 4 & 5 \end{vmatrix} & \begin{vmatrix} 2 & 1 \\ 3 & 5 \end{vmatrix} & \begin{vmatrix} 2 & 1 \\ 3 & 4 \end{vmatrix} \\ \begin{vmatrix} -2 & 2 \\ 4 & 5 \end{vmatrix} & \begin{vmatrix} -1 & 2 \\ 3 & 5 \end{vmatrix} & \begin{vmatrix} -1 & -2 \\ 3 & 4 \end{vmatrix} \\ \begin{vmatrix} -2 & 2 \\ 1 & 1 \end{vmatrix} & \begin{vmatrix} -1 & 2 \\ 2 & 1 \end{vmatrix} & \begin{vmatrix} -1 & -2 \\ 2 & 1 \end{vmatrix} \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 7 & 5 \\ -18 & -11 & 2 \\ -4 & -5 & 3 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & -7 & 5 \\ 18 & -11 & -2 \\ -4 & 5 & 3 \end{bmatrix}$$

cofactor matrix

$$\det(C) = \begin{vmatrix} -1 & -2 & 2 & -1 & -2 \\ 2 & 1 & 1 & 2 & 1 \\ 3 & 4 & 5 & 3 & 4 \end{vmatrix} = -5 - 6 + 16 - (-20 - 4 + 6) = 23$$

$6 \quad -4 \quad -20 \quad -5 \quad -6 \quad 16$

$$C^{-1} = \frac{1}{\det(C)} \text{adj}(C) = \frac{1}{23} \begin{bmatrix} 1 & 18 & -4 \\ -7 & -11 & 5 \\ 5 & -2 & 3 \end{bmatrix}$$

transpose of cofactor matrix